Multiply Decimals

Lesson 1-6

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

4 × 7 = 28, so the product is 28

Product

The answer when you multiply.

In 3.25, the decimal point separates 3 (ones) from 25 (tenths and hundredths)

Decimal point

The dot that splits the whole number from the part after it.

3.7 × 4.2: round to 4 × 4 = 16, so the exact answer should be near 16

Estimate

A close answer you get by rounding first.

In 0.36: the 3 is in the tenths place (0.3) and the 6 is in the hundredths place (0.06)

Place value

What a digit is worth based on where it sits in a number.

3.4 (1 place) × 2.6 (1 place) = 8.84 (2 places total)

Decimal places

How many digits come after the decimal point.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can multiply decimals and place the decimal point correctly in the product.

1️⃣ Notice

The station needs 4.5 meters of heat shielding that costs $12.60 per meter.

2️⃣ Key idea

I can multiply decimals and place the decimal point correctly in the product.

3️⃣ Words I'll use

Product **Decimal point** **Estimate** **Place value** **Decimal places**

🤔 Think About It

- What two quantities are being multiplied?
- How many total decimal places do the two factors have?
- How can estimating help you check if your answer is reasonable?

👀 Watch

Watch your teacher model one example. Jot what you see.

____

🤝 We try

Solve the next one together as a class.

____

____

✏️ You try

Now try one on your own.

____

____

Turn & Talk — Discussion Points

Level 1 support · Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The station needs 4.5 meters of heat shielding at $12.60 per meter. About how much will it cost, and how did you estimate before multiplying?

Level 1 support

Start here: Rounding the factors gives a reasonable estimate of the product.

💡 **Need a hint?**

Sentence starters (optional)

I estimate about $____ because 4.5 is close to ____ and $12.60 is close to ____.

Estimo unos $____ porque 4.5 está cerca de ____ y $12.60 está cerca de ____.

So the real answer should be near ____.

Entonces la respuesta real debe estar cerca de ____.

Word bank: product · decimal point · estimate · place value · decimal places

Listen for: *Students round to about 5 × $13 = $65 (or 4 × $13) to predict the cost is in the $50-$65 range, naming 4.5 and 12.60 as the factors.*

Level 2

Will the exact product be larger or smaller than your estimate of 5 × $13? Justify using how you rounded each factor.

- The exact product is ____ than 5 × $13 because ____.
- I rounded 4.5 ____ and 12.60 ____, so ____.

EXPLORE

When you multiplied 4.5 × 12.60, how did you decide where to put the decimal point in the product?

Level 1 support

Start here: Count the total decimal places in both factors to place the point.

💡 **Need a hint?**

Sentence starters (optional)

I counted ___ decimal places in the factors.

Conté ___ lugares decimales en los factores.

So my product has ___ decimal places, giving $___.

Entonces mi producto tiene ___ lugares decimales, dando $___.

Word bank: product | decimal point | estimate | place value | decimal places

Listen for: *Students count the decimal places, multiply the digits, then place the point to get $56.70, and use the estimate to confirm placement.*

Level 2

Use your launch estimate to argue that $56.70 (not $5.67 or $567) is where the decimal point belongs. Explain.

- The answer must be $56.70 because my estimate was about ___.
- It cannot be $5.67 or $567 because ___.

CONNECT

A family buys 3.5 pounds of deli meat at $6.80 per pound. How do you multiply the decimals to find the total cost, and how can estimating check it?

Level 1 support

Start here: Multiply the numbers, count decimal places, then check with an estimate.

💡 **Need a hint?**

Sentence starters (optional)

The total cost is $___ because I multiplied ___ × ___.

El costo total es $___ porque multipliqué ___ × ___.

My estimate of about $___ shows the decimal point is in the right place.

Mi estimación de unos $___ muestra que el punto decimal está en el lugar correcto.

Word bank: **product** **decimal point** **estimate** **place value** **decimal places**

Listen for: *Students compute 3.5 × 6.80 = 23.80, estimate about 4 × $7 = $28 to confirm, and explain counting decimal places to place the point.*

Level 2

Why does multiplying by a number less than 1 (like 0.5 pound) give a product SMALLER than the price per pound? Explain with this example.

- Multiplying by less than 1 makes the product smaller because ___.
- For 0.5 pound the cost would be ___, which is ___ than $6.80.

REFLECT

How is placing the decimal point in a product different from lining up decimals when adding? Why don't you line up the points to multiply?

Level 1 support

Start here: For products you count decimal places; for sums you align the points.

💡 **Need a hint?**

Sentence starters (optional)

To multiply I ____ the decimal places, but to add I ____ the decimal points.

Para multiplicar ____ los lugares decimales, pero para sumar ____ los puntos decimales.

I do not line up points to multiply because ____.

No alineo los puntos para multiplicar porque ____.

Word bank: product | decimal point | estimate | place value | decimal places

Listen for: *Students contrast multiplication (multiply digits, then count total decimal places) with addition (align decimal points first) and explain why the rules differ.*

Level 2

Predict the number of decimal places in 0.25 × 0.4 without multiplying, then check. Explain how you knew.

- I predict ____ decimal places because ____.
- After multiplying I got ____, which ____ my prediction.

Guided Examples

Example 1

What is 3.4 × 2.6?

Solution: 34 × 26 = 884. There are 2 total decimal places (1 + 1), so place the decimal to get 8.84.

Answer: A. 8.84

Example 2

What is 0.7 × 1.5?

Solution: 7 × 15 = 105. There are 2 total decimal places (1 + 1), so place the decimal to get 1.05.

Answer: A. 1.05

Example 3

What is 5.3 × 4?

Solution: 53 × 4 = 212. There is 1 total decimal place, so place the decimal to get 21.2.

Answer: A. 21.2

Write About the Math

The Writing Revolution

I can explain my work using the words product, decimal point, estimate, and decimal places.

1. Kernel Sentence

subject + verb

Model: Decimal point is the dot that splits the whole number from the part after it.
Punto decimal es el punto que separa el número entero de la parte que sigue.

Write a kernel sentence about decimal point. Use a subject and a verb.
Escribe una oración base sobre punto decimal. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Decimal point matters in math
Punto decimal importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because / ***porque***

Decimal point matters in math because ____.
Punto decimal importa en matemáticas porque ____.

____

but / ***pero***

Decimal point matters in math, but ____.
Punto decimal importa en matemáticas, pero ____.

____

so / ***entonces***

Decimal point matters in math, so ____.
Punto decimal importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about decimal point.
Di un hecho verdadero sobre decimal point.

Decimal point ____.

Question
Pregunta

Ask a question about decimal point.
Haz una pregunta sobre decimal point.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about decimal point.
Muestra entusiasmo sobre decimal point.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with decimal point.
Dile a un compañero qué hacer con decimal point.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I estimate about $____ because 4.5 is close to ____ and $12.60 is close to ____.
Estimo unos $____ porque 4.5 está cerca de ____ y $12.60 está cerca de ____.

So the real answer should be near ____.
Entonces la respuesta real debe estar cerca de ____.

I counted ____ decimal places in the factors.
Conté ____ lugares decimales en los factores.

Try It

Solve on your own. Check the answer key when you are done.

1. Which estimate is closest to 4.8×3.2?

A. $5 \times 3 = 15$

B. $4 \times 3 = 12$

C. $5 \times 4 = 20$

D. $48 \times 32 = 1{,}536$

Show your work:

2. A car travels 55.5 miles per hour for 2.4 hours. How far does it go?

A. 133.2 miles

B. 13.32 miles

C. 1,332 miles

D. 57.9 miles

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A family buys 3.5 pounds of deli meat at $6.80 per pound. Show how to find the total cost. Use estimation to check if your answer is reasonable.

Sentence starter: The total cost is $____ because 3.5 × $6.80 = ____. I can estimate: ____ × ____ = ____, which is close to my exact answer, so it is reasonable.

Show your work:

____

____

____

____

Reflect — Exit Ticket

What is 4.6 × 2.3?

A. 10.58

B. 105.8

C. 1.058

D. 10.48

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **Try It 1:** A. 5 × 3 = 15 *— Round 4.8 to 5 and 3.2 to 3. The estimate 5 × 3 = 15 is closest to the exact answer 15.36.*
2. **Try It 2:** A. 133.2 miles *— 555 × 24 = 13,320. Total decimal places: 1 + 1 = 2. Answer: 133.20 = 133.2 miles.*
3. **Exit Ticket:** A. 10.58 *— 46 × 23 = 1,058. There are 2 total decimal places (1 + 1), so the answer is 10.58.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Decimal point is the dot that splits the whole number from the part after it.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).